

DUOLUN, China

WMO: 542080

Lat: 42.1939N

Lon: 116.4664E

Elev: 4091

StdP: 12.65

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.4	-13.4	-25.3	1.6	-11.8	-21.6	1.9	-10.2	26.3	9.8	23.4	7.2	4.0	270	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	85.0	61.9	81.9	61.0	79.2	60.5	68.2	77.7	66.4	75.0	64.8	73.2	7.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	108.6	72.6	63.6	102.4	71.1	61.9	96.3	69.5	35.1	77.8	33.6	75.2	32.2	73.2	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.3	20.1	17.4	DB	-24.3	89.8	5.6	3.7	-28.4	92.5	-31.7	94.7	-34.8	96.8	-38.9	99.5	(4)
(5)			WB	-24.6	70.2	5.6	2.0	-28.6	71.6	-31.9	72.8	-35.0	73.9	-39.0	75.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.1	3.3	11.4	26.4	41.7	53.4	62.3	67.7	64.3	54.3	40.2	22.7	8.0
	DBStd	23.58	9.05	10.32	10.18	9.02	7.53	5.49	4.45	4.85	6.76	8.43	10.48	8.91
	HDD50	6060	1447	1082	733	276	50	1	0	0	35	317	818	1301
	HDD65	10009	1912	1502	1197	700	367	114	21	70	324	768	1268	1766
	CDD50	1722	0	0	1	27	155	370	547	444	164	14	0	0
	CDD65	196	0	0	0	0	7	33	104	49	3	0	0	0
	CDH74	2244	0	0	0	13	189	484	948	525	84	1	0	0
	CDH80	493	0	0	0	0	39	103	244	97	10	0	0	0
Wind	WSAvg	6.9	7.0	7.1	8.7	9.0	8.4	6.1	5.1	4.6	5.4	7.0	7.4	7.5
Precipitation	PrecAvg	14.4	0.1	0.1	0.3	0.6	1.1	2.4	4.1	3.4	1.5	0.6	0.2	0.1
	PrecMax	20.2	0.5	0.5	1.0	3.0	2.7	6.1	7.0	9.6	3.3	2.1	0.6	0.3
	PrecMin	10.2	0.0	0.0	0.0	0.0	0.1	0.9	0.5	0.9	0.5	0.0	0.0	0.0
	PrecStd	2.8	0.1	0.1	0.2	0.6	0.6	1.3	1.7	1.8	0.8	0.5	0.2	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	44.5	60.9	75.9	84.5	87.0	90.4	87.1	81.0	70.4	54.1	36.7
		MCWB	25.3	32.6	41.6	50.2	55.3	58.8	66.1	63.6	57.7	50.7	39.1	27.7
	2%	DB	26.9	38.3	53.9	69.9	79.1	83.0	85.9	82.5	76.2	64.9	48.1	30.7
		MCWB	21.1	28.9	37.7	46.9	53.3	58.9	64.4	62.9	56.3	47.9	36.5	24.4
	5%	DB	22.5	33.6	48.5	64.6	74.8	79.5	82.3	79.5	72.6	60.3	42.9	26.1
		MCWB	18.0	26.1	34.9	44.3	51.4	58.3	63.5	61.8	54.4	45.1	33.9	21.2
	10%	DB	18.4	29.1	43.5	59.6	70.5	76.2	79.4	76.6	69.1	55.8	38.4	22.4
		MCWB	14.9	23.1	32.0	42.3	50.1	57.4	63.2	61.4	53.5	43.1	31.3	18.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.5	33.6	42.7	52.9	59.0	65.7	70.8	69.8	63.0	53.1	41.2	28.8
		MCDB	32.3	42.0	58.4	68.7	74.1	76.6	81.7	79.6	72.7	64.6	49.9	36.2
	2%	WB	21.3	29.7	38.8	48.3	56.0	63.3	68.9	67.6	59.9	49.8	37.2	24.7
		MCDB	26.5	37.6	52.2	65.2	72.5	74.4	79.1	76.5	69.1	60.9	46.8	30.3
	5%	WB	18.1	26.3	35.5	45.9	54.0	61.6	67.4	65.8	57.5	47.0	34.4	21.4
		MCDB	22.3	33.5	47.3	61.7	68.6	72.6	76.5	73.8	67.5	57.1	42.3	26.1
	10%	WB	14.9	23.0	32.5	43.3	51.9	60.1	65.8	63.9	55.5	44.4	31.3	18.4
		MCDB	18.4	29.0	43.1	57.5	65.6	71.0	74.3	71.6	65.7	53.4	38.3	22.3
Mean Daily Temperature Range	5% DB	MDBR	20.0	22.5	22.4	23.7	25.0	22.9	20.6	21.8	24.1	22.7	20.0	18.9
		MCDBR	24.8	27.7	29.3	32.7	33.1	29.6	27.1	27.2	30.2	30.3	25.2	24.2
		MCWBR	20.2	20.1	17.2	15.5	13.4	10.7	9.5	10.7	12.7	15.5	15.6	19.2
	5% WB	MCDBR	24.7	27.1	27.8	28.7	27.1	22.5	20.6	20.7	24.2	26.3	23.7	23.8
		MCWBR	20.2	19.8	16.6	14.7	11.7	9.6	8.7	9.5	11.6	14.6	15.1	19.0
Clear-Sky Solar Irradiance	taub	0.275	0.326	0.368	0.427	0.482	0.533	0.495	0.475	0.429	0.374	0.304	0.284	
	taud	2.312	2.162	2.076	1.973	1.879	1.830	1.991	2.014	2.063	2.139	2.281	2.284	
	Ebn at Noon	276	274	275	266	254	241	249	249	252	252	260	259	
	Edh at Noon	27	37	46	55	62	65	55	52	46	38	28	25	
All-Sky Solar Radiation	RadAvg	779	1112	1541	1820	2039	1927	1863	1760	1507	1154	808	658	
	RadStd	62	86	111	136	124	141	94	95	82	70	61	52	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	-2.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	-2.29	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page